Examiner only

8. Newton's laws describe the motion of objects.

(a) (i) State what quantity is an expression of the inertia of a body. [1]

....

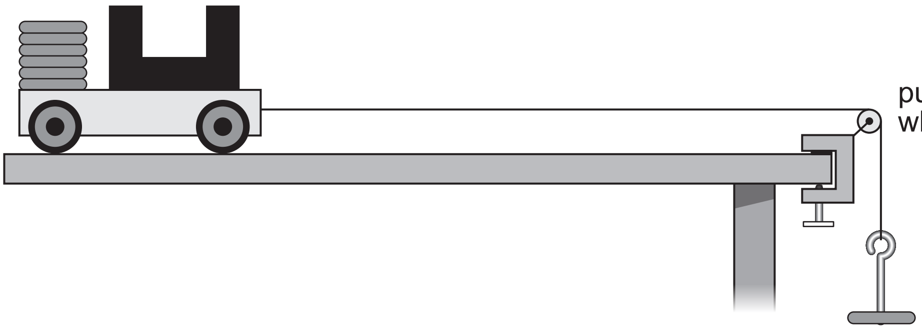
(ii) State Newton's 1st law. [2]

....

....

....

(b) Students investigate Newton's 2nd law.
They measure the acceleration of a trolley for different forces using light gates and a datalogger.
The method they follow is given below.



Method

1. Attach a 50 g mass hanger to the string. This gives an applied force of 0.5 N.
2. Release the trolley and record the acceleration from the datalogger.
3. Repeat step 2 twice more to give 3 results.
4. Remove one of the 50 g slotted masses from the trolley and place it on the mass hanger to increase the applied force by 0.5 N.
5. Repeat steps 2 to 4 until all the slotted masses have been placed on the mass hanger.

(i) State the independent variable. [1]

....

(ii) State the dependent variable. [1]

....

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(iii) The same trolley is used throughout the experiment.
State **one** other controlled variable. [1]

....

(iv) Some of the students' results are shown below.

Applied force (N)	Acceleration (m/s^2)			
	Trial 1	Trial 2	Trial 3	Mean
0.5	0.299	0.323	0.309	0.310
1.0	0.618	0.619	0.621	0.619
1.5	0.923	0.936	0.934	0.931

Sara states that the results prove Newton's 2nd law because the acceleration is proportional to the applied force.
Explain whether you agree. [2]

....

....

....

(v) I. Use the results for an applied force of 0.5 N and an equation from page 2 to calculate the total mass of the trolley and the slotted masses. [3]

mass = ... kg

II. Explain how the students could check the accuracy of this value of the mass. [1]

....

....

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END OF PAPER